CONSTRUCTOR UNIVERSITY

Bug World
Task Sheet

Software Engineering Project

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Overview. The Bug World program system implements a dual-player tournament where each player controls one bug cohort (black or red) moving in some synthetic world. The bugs are controlled via a special purpose robot language. The simulation proceeds in "ticks". At each tick, the next instruction of red and black cohort each is evaluated, and the program counter is advanced by 1. When a program has executed its last instruction, the program counter is reset to start.

Technical details. The simulator (**sim**) is given as an executable which can be launched standalone. It communicates via two ports for reading red and black commands, respectively.

A bug program accepted by the simulator is a binary data stream; for convenience, an assembler language is defined and the corresponding assembler program (**asm**) is available which generates the binary program from ASCII assembler code provided. The protocol used for forwarding bug programs to **sim** is documented.

Executables have been generated using gcc version 13.3.0 under Ubuntu 24.04.3.

Files. All material is available on the working machine under
https://teaching.peter-baumann.org/ (ignore certificate complaint)

You find the executables **sim** and **asm**, documentation, sample bug worlds, and sample bug programs in these directories:

- docs/ -- documentation
- bin/ -- **asm**, **sim**
- worlds/ -- sample world maps
- bugs/ -- sample bug programs

Task. A visual frontend to **sim** is to be implemented in C++ showing the state of the map at each tick.

A bug program has to be developed in the last sprint; on the bugs submitted a tournament is done where the winner ranking determines grades for that sprint. Note that **sim** can change over the semester to convey a changed simulation behavior.

Teams need to significantly advance the code base (functionality, documentation, tests, quality) in each sprint. Use of AI, when discovered, impacts the team sprint assessment negatively.